P.7 A Library of Parent Functions

Name _____ Period ____ Date _____

Pre-Calculus P.7 · Student sheet

I can…

- write a linear function from two values.
- identify the common parent functions.
- evaluate and graph step and piecewise functions.

Vocabulary · fill in the missing word

Parent function — the _____ function of a family, like $f(x) = x^2$ for quadratics

Linear function — $f(x) =$ _____

Greatest integer function — $[\![x]\!]$ = the greatest integer less than or _____ to x

Part 1 · Match the graphs

Match each graph to its parent function and write one feature.

Name	Graph #	Feature
Absolute value		
Square root		
Squaring		
Cubic		
Reciprocal		
Greatest integer		

Part 2 · Notes and examples (fill in each note, then work the example)

Linear functions: $f(x) = mx + b$; find m from two points, then find _____.

Example 1

Write the linear function with $f(1) = 3$ and $f(4) = 0$.

Parent functions to know: constant, identity, absolute value, square root, squaring, cubic, reciprocal, and _____.

Example 2

Name the parent function of each: $y = x^2$, $y = \sqrt{x}$, $y = |x|$, $y = \frac{1}{x}$, $y = x^3$.

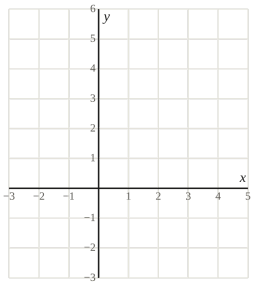
Greatest integer: $[\![x]\!]$ is the greatest integer _____ x.

Example 3

Evaluate $[\![-1]\!]$, $[\![-1.5]\!]$, $[\![1.5]\!]$, and $[\![0.1]\!]$.

Example 4

$f(x) = 2x + 3$ if $x \le 1$; $-x + 4$ if $x > 1$. Find $f(1)$ and $f(3)$, and sketch.



Example 5

A garage charges \$3 for each hour or part of an hour. How much is parking for 2.5 hours?

Part 3 · Partner practice

Partner 1

Write the linear function with $f(-2) = 6$ and $f(4) = -3$.

Partner 2

Evaluate $[\![3.9]\!]$ and $[\![-3.9]\!]$.